

Dengue Weekly Hotspot & Risk Report

Target Week: 52, 2025

Area Covered: Kuala Lumpur, Malaysia

Generated By: Dengue Prediction System

Date Generated: 2025-12-12

2. Executive Summary

For Week 52, a total of **45** hotspot grids were identified across the scanned area. The total accumulated risk score is **92.38**, with an average grid risk of **2.05**. The highest recorded risk intensity in a single grid is **2.46**.

The report highlights critical clusters that require immediate public health attention.

3. Data Sources & Methodology

Historical Data Source: *ultimate_combined_data.csv* containing past confirmed dengue cases.

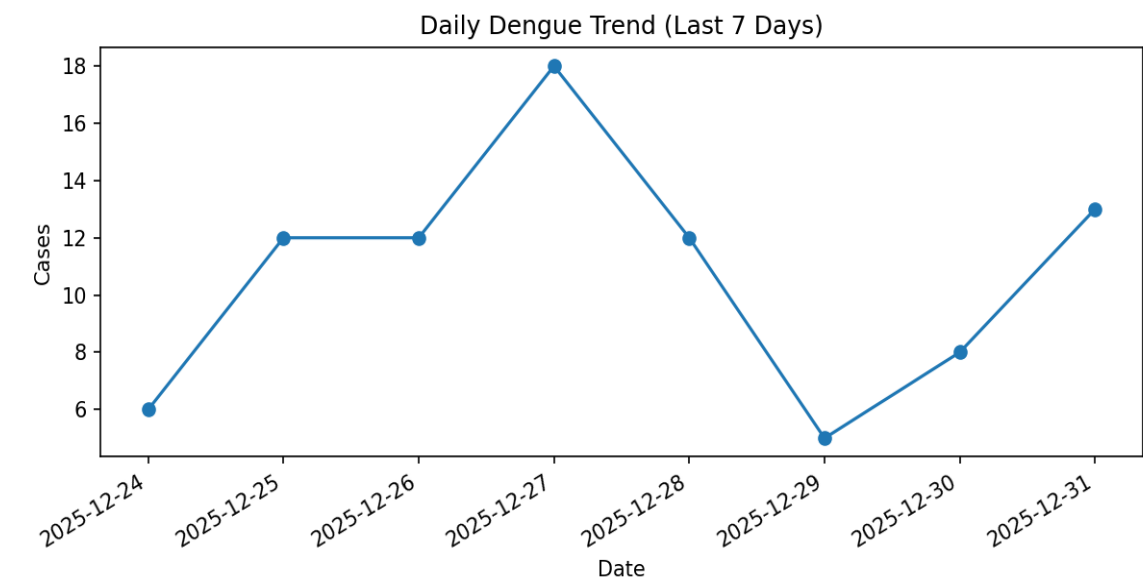
Prediction Source: Machine-learning generated grid risk (*heatmap_data.json*).

Hotspot Definition: Any grid point with a predicted risk probability > 0 (inclusive of all detected risks).

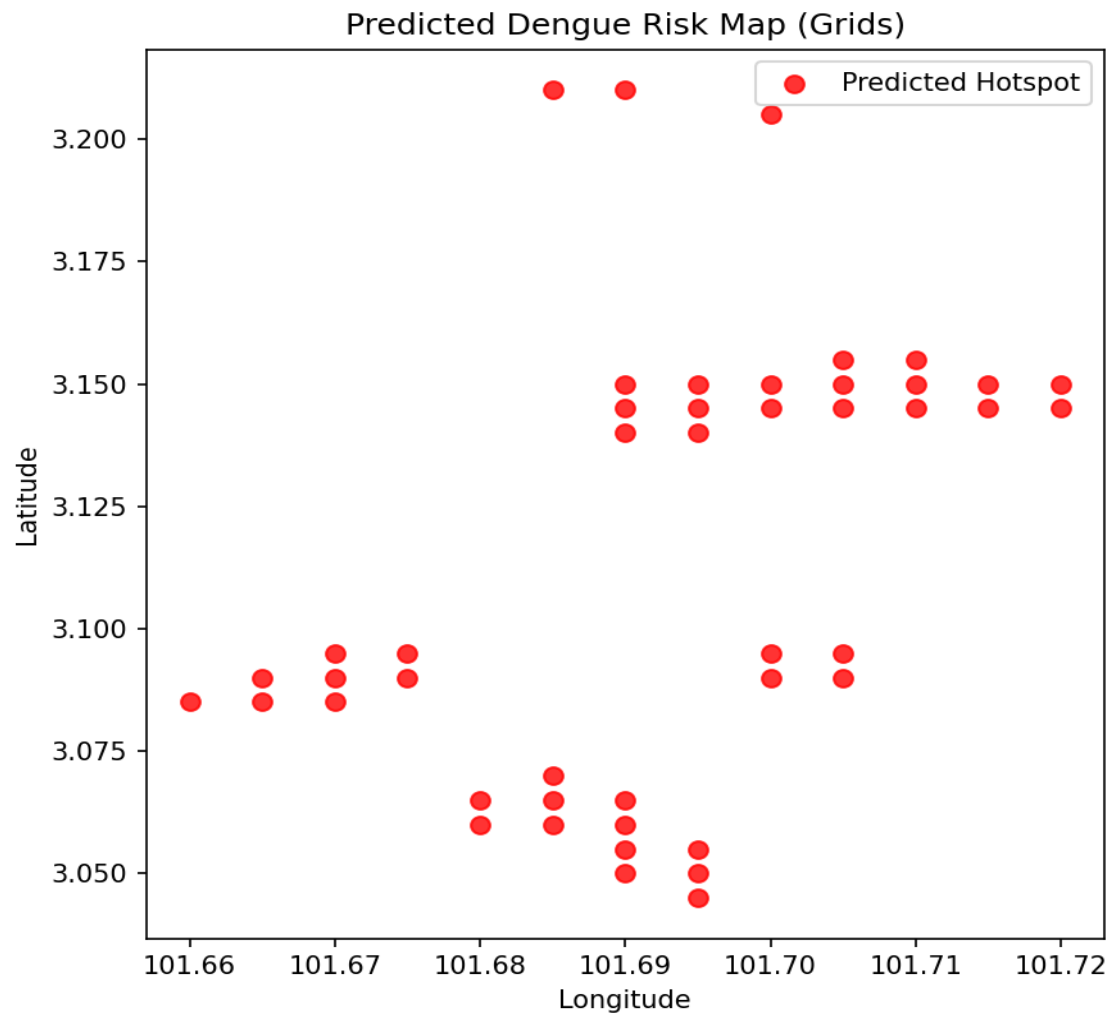
Spatial Resolution: ~500m Grid-based latitude/longitude cells.

Time Resolution: ISO Weekly predictions.

4. Historical Dengue Trend



5. Predicted Spatial Risk Map



6. Hotspot Summary Table

#	Location Area	Peak Risk	Risk Level	Grids
1	Bukit Jalil (Vista Komanwel)	2.46	MEDIUM	12
2	Desa Petaling	2.02	MEDIUM	4
3	Changkat Bukit Bintang	2.00	MEDIUM	11
4	No. 51, Jalan Alor, 50200 Kuala Lumpur	2.00	MEDIUM	1
5	Old Klang Road (Pearl Point)	2.00	MEDIUM	8
6	Pavilion KL (Residences)	2.00	MEDIUM	6
7	Ppr Batu Muda	2.00	MEDIUM	3

7. Statistical Summary

Total Predicted Risk: 92.38
Mean Risk: 2.05
Standard Deviation: 0.15
Max Risk Score: 2.46

8. Comparison with Previous Weeks

Comparative analysis with previous week's prediction is currently unavailable in this version.

9. Risk Interpretation & Implications

Environmental Factors: High risk is often correlated with urban density and recent rainfall patterns.

Implications: Areas flagged as "High" or "Medium" risk require immediate attention for mosquito breeding site inspections.

Surveillance: Public health teams should prioritize these clusters for early intervention.

10. Recommendations

For Authorities:

- Conduct targeted fogging in the identified hotspot grids.
- Increase surveillance and larviciding in emerging danger zones.
- Launch localized public awareness campaigns.

For Public:

- Inspect home compounds for stagnant water (mosquito breeding sites).
- Use mosquito repellent and wear long-sleeved clothing.
- Seek early medical attention if experiencing fever or joint pain.

11. Limitations & Assumptions

- **Prediction Model:** Depends on historical patterns; sudden environmental shifts may not be fully captured.
- **Risk $\neq$ Cases:** A high risk score indicates probability, not a guaranteed outbreak.
- **Approximation:** Environmental variables are approximated for grid locations.
- **Human Movement:** The model does not currently account for daily human mobility patterns.

12. Appendix

Weight: The raw probability score from the machine learning model (scaled 0-10).

Grid Resolution: The map is divided into fine-grained cells to detect micro-clusters.

Model Accuracy: Based on training validation against historical outbreak data.